University of Salerno

MSc in Computer Engineering
Department of Information and Electrical Engineering and Applied Mathematics
(DIEM)

Solar Filament Image Segmentation Using a U-Net Architecture

Machine Learning project work

Team

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1 Introduction

Automatic solar-filament segmentation requires detecting thin, low-contrast structures in full-disc images. The task is complicated by the strong imbalance between foreground and background, the variation in filament shape and size, and the availability of multiple annotations for the same observation. A useful prediction must therefore identify filament pixels without producing too many spurious or fragmented regions.

This work develops a reproducible pipeline for image segmentation and subsequent instance extraction. Its main model is a U-Net with a pre-trained ResNet18 encoder, chosen to combine semantic information with spatial detail. Mean Dice is the primary metric, while Panoptic Quality (PQ) also evaluates instance recognition. GPU-memory use and execution time are measured because the solution must be trainable and usable with limited resources.

During development, **22 experimental configurations were run**. The report focuses on the eight numbered experiments that determined the main design choices. It also compares the full-resolution U1 and U2 variants and the three YOLO detector experiments. This organisation reconstructs the path from the baseline to the delivered model while also showing alternatives that improved PQ at a higher computational cost.

The following sections define the metrics and validation protocol, describe the architecture and training procedure, and discuss the most significant experiments. The final part analyses prediction errors, computational cost and the main limitations of the solution.

2 Evaluation criteria

2.1 Mean Dice

For validation record i, $\Omega_i = \{1, \ldots, H_i\} \times \{1, \ldots, W_i\}$ is the set of pixel coordinates of an image with height H_i and width W_i, that is, its discrete domain. Let $P_i \subseteq \Omega_i$ be the predicted foreground mask and $G_i \subseteq \Omega_i$ the union of its annotated instances. The Dice coefficient is

$$D_i = \frac{2|P_i \cap G_i| + \varepsilon}{|P_i| + |G_i| + \varepsilon}, \tag{1}$$

where ε avoids division by zero. If both masks are empty, the implementation assigns $D_i = 1$. Over the N annotator-image records in the validation set, the reported metric is

$$\text{Mean Dice} = \frac{1}{N} \sum_{i=1}^{N} D_i. \tag{2}$$

This union-mask measure is appropriate for the strong class imbalance because it does not reward correctly classified background pixels. However, it does not describe how the foreground is divided into separate objects: a filament split into several components can still retain substantial pixel overlap.

2.2 Panoptic Quality

PQ is therefore used as a complementary instance-level metric. For every annotator-image record, predicted and annotated instances are associated when their Intersection over Union is greater than 0.5. With this threshold, valid associations are one-to-one. True matches and unmatched predicted or annotated instances are accumulated over the validation split as TP, FP and FN. The metric is

$$\text{PQ} = \underbrace{\frac{\sum_{(p,g)\in\text{TP}} \text{IoU}(p, g)}{|\text{TP}|}}_{\text{SQ}} \times \underbrace{\frac{|\text{TP}|}{|\text{TP}| + \frac{1}{2}|\text{FP}| + \frac{1}{2}|\text{FN}|}}_{\text{RQ}}. \tag{3}$$

Segmentation Quality (SQ) measures the mean overlap of matched instances, whereas Recognition Quality (RQ) penalises missing and spurious instances. The distinction is relevant in this study because configurations with similar Mean Dice can produce different numbers of fragmented or false-positive components.

The two measures are reported for one explicitly defined operating point at a time. In particular, the best Dice from one output configuration is never combined with the best PQ from another.

3 Data organisation and validation

The labelled data contain 707 different observations but 1154 annotation records. Several observations were annotated by two or three experts. Randomly splitting the individual records could therefore assign annotations of the same image to both training and validation. In that case, the model would be evaluated on images already seen during training, leading to an overestimate of its performance. To prevent this form of data leakage, I performed the split at file level and assigned all annotations of the same observation to a single subset.

The final split uses seed 22 and contains 601 training files and 106 validation files, corresponding to 985 and 169 annotation records. The separate Kaggle test set is used only by the inference notebook. The exact membership is stored in `split_seed22.csv`, so the reported validation set can be reconstructed.

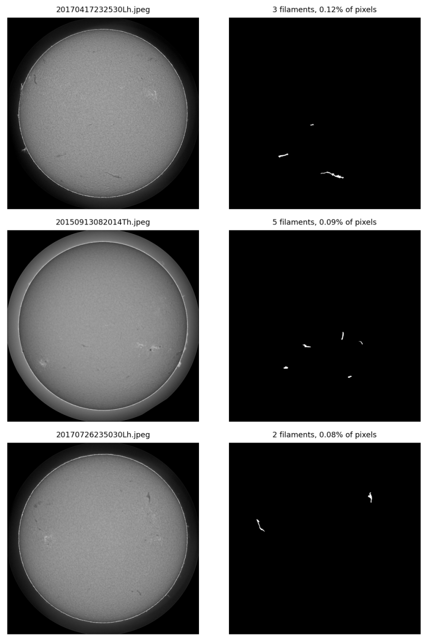


Figure 1: Examples used to inspect image contrast and annotation variability.

The masks contain 8199 filament polygons and foreground covers only about 0.4% of an image. The median native instance area is roughly 1230 pixels. On observations with multiple annotations, mean pairwise Dice is about 0.62. This does not define a human upper bound, but it shows that some apparent prediction errors may also reflect differences between annotators.

Validation was used for checkpoint selection, comparison between experiments and final output settings. Reusing one split in this way can make the final score optimistic, so small differences are treated with caution.

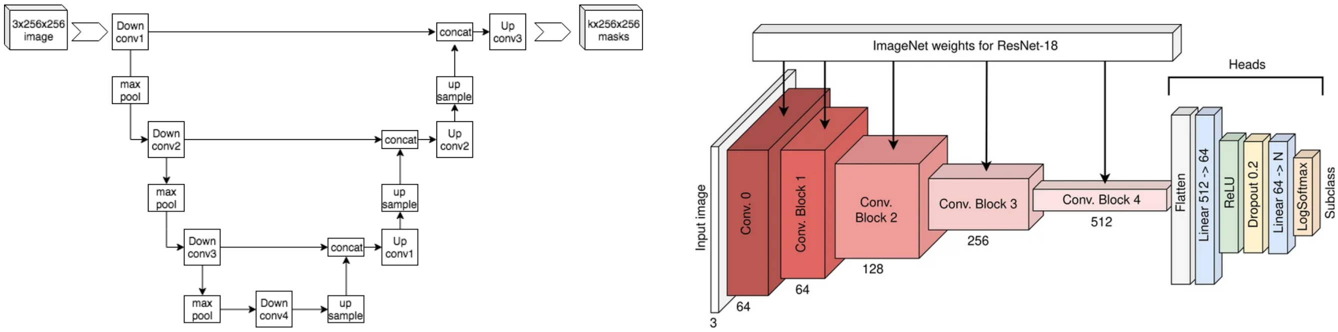
4 Architecture, preprocessing and training

4.1 Network architecture

The implemented network follows the U-Net encoder-decoder structure. Its contracting path extracts progressively more abstract features while reducing their spatial resolution; the expanding path restores the original resolution. Skip connections concatenate encoder features with decoder features at corresponding scales, allowing the decoder to recover localisation information that would otherwise be lost during downsampling.

A ResNet18 pre-trained on ImageNet replaces the original U-Net encoder. Residual blocks learn a transformation $F(x)$ and add the identity shortcut, producing $F(x) + x$; this facilitates gradient propagation through the feature extractor. Only the convolutional stem and the four residual stages are retained. The flattening and classification heads shown in Figure 2(b) are discarded, while feature maps with 64, 128, 256 and 512 channels are supplied to the U-Net decoder.

The reference diagrams are generic. In the implemented network, the first ResNet18 convolution is adapted from three RGB channels to one grayscale channel by combining the pre-trained weights. The decoder ends with a 1×1 convolution that returns one foreground logit per pixel. The complete network has 14.34 million trainable parameters.



(a) U-Net: encoder, decoder and multi-scale skip connections.

(b) ResNet18: residual feature hierarchy and original classification heads.

Figure 2: Reference schemes for the two components combined in the implemented model.

4.2 Input and inference pipeline

The selected experiment preserves the native 2048^2 sampling. During training, a random 1536^2 crop is extracted from each frame. During inference, four overlapping 1536^2 tiles cover the complete image; their probability maps are blended before thresholding. This procedure keeps native pixel detail while bounding the memory used by each forward pass.

$$\boxed{\text{native grayscale frame}} \longrightarrow \boxed{1536^2 \text{ train crop / four inference tiles}} \longrightarrow \boxed{\text{ResNet18 U-Net}}$$
$$\longrightarrow \boxed{\text{blended } 2048^2 \text{ probabilities}} \longrightarrow \boxed{\text{components}} \longrightarrow \boxed{\text{RLE submission}}$$

Images and masks are decoded once into uint8 memory-mapped files. The cache avoids repeating JPEG decoding and polygon rasterisation at every epoch without storing the complete dataset in RAM. It contains only unmodified samples; augmentation is generated independently at each epoch. During training I apply a random 1536^2 crop, horizontal and vertical flips (probability 0.5 each), a random rotation of 0, 90, 180 or 270 degrees and photometric perturbations. Brightness and contrast vary within $\pm 20\%$, and Gaussian noise is added in 30% of the cases. Geometric transforms are applied to the image and mask together.

4.3 Optimisation

For one sample, let z_j be the logit at pixel j, $p_j = \sigma(z_j)$ its foreground probability and $y_j \in \{0, 1\}$ the target, over M pixels. The loss terms are:

- **Binary Cross-Entropy (BCE):** measures the difference between the predicted probability and the target for each pixel, providing local pixel-wise supervision.
- **Soft Dice loss:** measures the overlap between predicted and ground-truth masks. It is useful under strong foreground-background imbalance because it directly considers the area of the two regions.
- **Combined loss:** averages the two terms, assigning weight 0.5 to each.

The definitions used are

$$\begin{aligned} \mathcal{L}_{\text{BCE}} &= -\frac{1}{M} \sum_{j=1}^{M} \left[y_j \log(p_j) + (1 - y_j) \log(1 - p_j) \right], \\ \mathcal{L}_{\text{Dice}} &= 1 - \frac{2 \sum_{j=1}^{M} p_j y_j + \varepsilon}{\sum_{j=1}^{M} p_j + \sum_{j=1}^{M} y_j + \varepsilon}, \quad \varepsilon = 10^{-6}, \\ \mathcal{L} &= 0.5\, \mathcal{L}_{\text{BCE}} + 0.5\, \mathcal{L}_{\text{Dice}}. \end{aligned} \tag{4}$$

The stabilising constant ε avoids a zero denominator, and both terms are averaged over the mini-batch. No positive-class weight is used in experiment 13. AdamW is configured with learning rate 10^{-4}, weight decay 10^{-4} and cosine decay over at most 20 epochs. Checkpoint selection and early stopping are based on validation Dice; the saved checkpoint is epoch 12.

The effective batch size is 2. One crop is processed at a time and gradients are accumulated over two forward/backward passes. Gradient checkpointing and bfloat16 mixed precision are also enabled. The training stopped after 15 epochs and required 3.04 GB at peak; inference required 2.53 GB.

Finally, the probability map is converted into connected components using the operating point selected on validation. The final values and the distinction between Dice- and PQ-oriented configurations are reported in Section 6.

5 Experimental strategy

5.1 Hypotheses and decision criteria

The experimental plan was defined around a limited set of hypotheses. Each hypothesis was associated with a measurable comparison and with a decision that could be taken from the result. Table 1 summarises this structure.

Table 1: Hypotheses used to guide the experimental work.

Hypothesis	Test	Decision criterion
Pre-trained features are useful on the limited labelled set.	Compare a U-Net trained from scratch with a ResNet18 encoder.	Retain transfer learning only if validation Dice improves.
Fragmentation is mainly a loss-function problem.	Introduce Tversky and clDice while keeping the low input resolution.	Retain the specialised loss only if both Dice and PQ improve.
Thin filaments are lost during downsampling.	Increase the input from 512 to 1024 and then 1536 pixels.	Prefer the scale that improves object recognition at acceptable cost.
Full-disk context and native detail produce different errors.	Compare resized full frames with native-resolution crops.	Evaluate Dice, PQ and the type of instance errors jointly.
High-resolution training can satisfy the memory targets.	Replace a physical batch of 2 with two accumulated micro-batches.	Accept the change if quality is retained and measured memory decreases.

During the study, 22 configurations were run. Section 6 reports only the eight numbered runs that provided evidence for the hypotheses above; the full-resolution U-Nets and YOLO detector experiments are compared separately in Section 7.2.

In the first transfer-learning run, the learning rate was also changed, while higher input resolutions required different memory settings. The results therefore support practical design decisions, but the entire score difference cannot always be attributed to one variable.

All final scores are recomputed at the native image resolution, and output parameters are selected only on validation data.

In some experiments, the training seed was changed a couple of times while keeping the split fixed. The resulting scores remained nearly unchanged, suggesting limited variability for the tested configurations.

5.2 Alternatives not retained

The main alternatives considered and then rejected were:

- **Alternative losses:** topology-oriented, soft and instance-balanced variants gave no stable gain.
- **Scale jitter:** inconsistent gains.
- **Post-processing:** detect-then-refine and component merging increased complexity or fusion risk.
- **ResNet34 encoder:** more memory without a Dice gain.
- **Detectors:** higher PQ, lower Dice and higher cost.

These results guided the choice of input scale and final architecture.

6 Experimental evidence and model selection

Table 2 reports the eight numbered configurations that most directly affected a design decision. Values marked with an asterisk belong to the initial evaluation protocol, in which foreground probabilities were converted into a binary mask using threshold 0.5; the remaining values were recomputed at native resolution using the stated validation objective.

Table 2: Numbered experiments retained in the decision path.

Exp.	Main comparison	Dice	PQ	Evidence obtained
1	U-Net from scratch, 512^2	0.6415*	–	reference baseline
2	pre-trained ResNet18 encoder	0.6552*	0.2505*	transfer is beneficial
3	Tversky + clDice, 512^2	0.4937*	0.0618*	loss hypothesis rejected
5	input resolution 512 → 1024	0.6665	0.4022	spatial scale is important
8	full image at 1536^2	0.6678	**0.4264**	strong single-model PQ
13	native image, 1536^2 crops	**0.6762**	0.3591	highest Mean Dice
15	1536^2, weighted BCE, batch 2	0.6769	0.4198	quality, but 5.20 GB
15m	run 15 with memory controls	0.6650	0.4153	memory-compliant alternative

6.1 From the baseline to spatial resolution

The main steps are summarised below:

- **Runs 1–2:** the pre-trained ResNet18 encoder provided a better starting point than random initialisation.
- **Run 3:** Tversky+clDice reduced both metrics at 512^2; fragmentation was therefore mainly linked to spatial scale.
- **Runs 5 and 8:** increasing the input to 1024^2 and 1536^2 improved PQ more than specialised losses, motivating the return to BCE+Dice.

6.2 Final candidates

Run 13 removed the full-frame resize and used 1536^2 crops from the native 2048^2 images. It achieved the highest Mean Dice, 0.6762, while remaining below the optional memory thresholds. It is therefore the model selected for delivery.

Runs 15 and 15m answer a different question. Weighted BCE on a complete resized frame reduced spurious instances and increased PQ. Run 15 required 5.20 GB during training; 15m reproduced the approach with gradient accumulation, checkpointing and mixed precision, reducing the peak to 2.63 GB. Compared with run 13, 15m gains 0.0562 PQ but loses 0.0112 Mean Dice. Since Mean Dice is the primary project metric, 15m is documented as the second final candidate rather than used as the delivered model. Its PQ-oriented submission obtained 0.35 on the public Kaggle leaderboard.

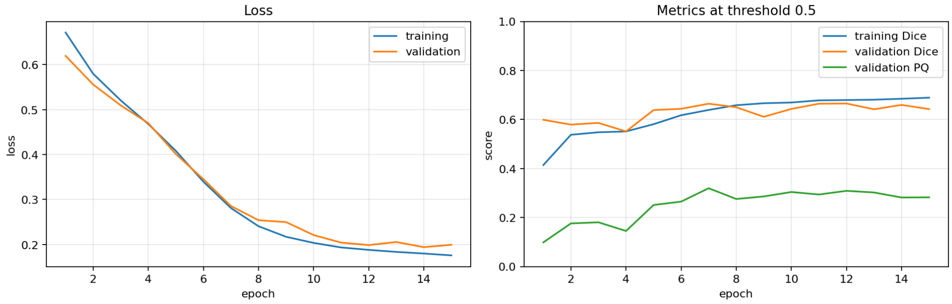


Figure 3: Training history of experiment 13. The saved Mean-Dice checkpoint is epoch 12; the final operating point is selected afterwards on validation probabilities.

6.3 Operating-point selection

The experiment-13 network returns one probability per pixel. Threshold, morphological closing, minimum component size and confidence gate were searched jointly at native resolution. The frozen Dice-oriented recipe is reported in Table 3.

Table 3: Inference recipe for the delivered model.

Setting	Selected value
checkpoint	experiment 13, epoch 12
probability threshold	0.30
closing iterations	0
minimum component size	128 pixels at 2048^2
confidence gate	disabled
validation Mean Dice	**0.6762**
validation PQ at the same point	0.3591

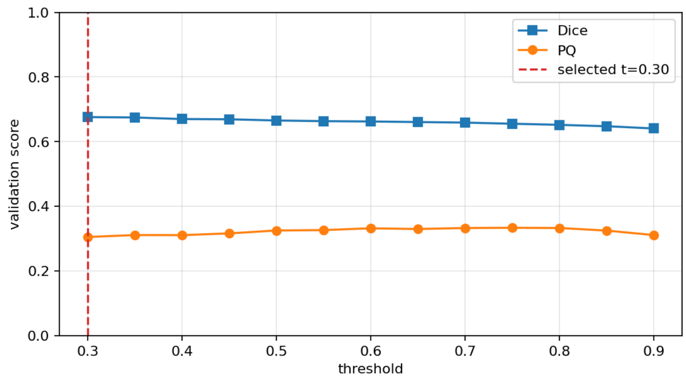


Figure 4: Validation Mean Dice and PQ favour different output settings for experiment 13.

The plot shows why the selection objective must be stated. Increasing component selectivity can improve PQ while reducing union-mask overlap. The reported Dice and PQ therefore always refer to the same frozen configuration.

7 Error analysis and additional models

7.1 Errors of the delivered model

At the selected operating point, experiment 13 produces 733 matched instances, 845 false positives and 411 false negatives on validation. Its SQ is 0.667 and its RQ is 0.539. The difference indicates that recognition errors have a larger effect on PQ than the boundaries of already matched objects.

Recall is strongly related to instance size: it is 0.44 below 500 pixels, 0.65 between 500 and 1000, and approximately 0.69–0.72 in the larger groups. Small or faint filaments are therefore the clearest weakness of the delivered model.

Table 4: Error categories for experiment 13 on validation.

	Category	Count	Meaning
FN (411)	pure miss	113	no predicted overlap
	near miss	245	overlap exists, but IoU ≤ 0.5
	split	53	one annotation covered by several pieces
FP (845)	no-overlap FP	418	no annotated pixel is touched
	unmatched	411	overlap exists, but IoU ≤ 0.5
	merge	16	one prediction overlaps several annotations

Figure 5: Experiment-13 validation examples: annotation, probability map and final connected components. A response outside one annotation can be a false positive or reflect disagreement between annotators.

The distributions in Figure 6 provide an additional view of the same result. Among overlapping ground-truth/prediction pairs, mean IoU is 0.516 and mean Dice is 0.648. These pairwise values do not replace the global metrics because they exclude components without a match.

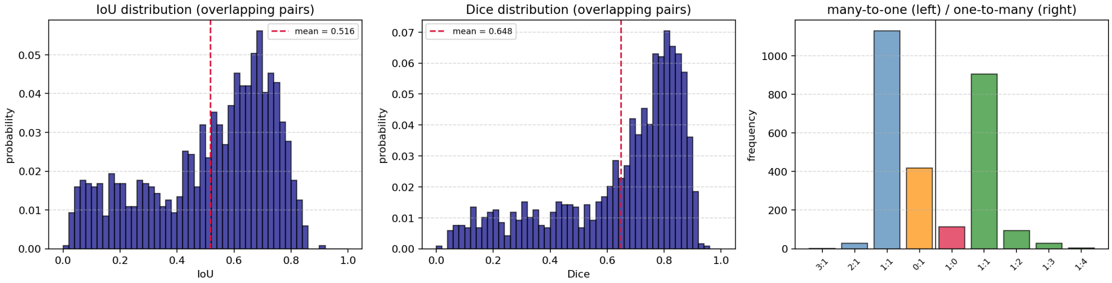


Figure 6: Pairwise overlaps and matching degrees for experiment 13 on validation.

7.2 Heavier models and the second final candidate

For completeness, Table 5 reports the configurations that explored greater capacity or a different balance between Mean Dice and PQ. Memory values are measured peaks, not estimates from parameter counts.

Table 5: Final candidate and higher-cost configurations. A dash indicates that no separate training or inference peak is applicable or available.

ID	Configuration	Dice	PQ	Train GB	Test GB
13	ResNet18 U-Net, native crops	**0.6762**	0.3591	3.04	2.53
15m	ResNet18 U-Net, full 1536^2	0.6650	0.4153	2.63	2.53
U1	ResNet18 U-Net, full 2048^2	0.6691	0.4211	9.20	4.32
U2	ResNet34 U-Net, full 2048^2	0.6670	0.4282	9.23	4.36
Y1	YOLOv8m-seg, 1792^2 (40 ep.)	0.6150	0.4170	21.59	–
Y2	YOLOv8m-seg, 1792^2 (39 ep.)	0.6181	0.4197	21.59	–
Y3	YOLOv8x-seg, 1792^2 (100 ep.)	0.6324	0.4263	38.74	–

U1 and U2 process the complete native-resolution frame. U2 replaces ResNet18 with ResNet34, increasing the trainable parameters from 14.34 to 24.45 million. Their PQ improves, but their training peaks exceed 9 GB and Mean Dice remains below experiment 13.

The YOLO experiments (Y1–Y3) achieved a reasonable PQ but lower Mean Dice than experiment 13. The largest model, Y3, reached 0.4263 PQ but required 38.74 GB of training memory; the detector branch was therefore not selected.

Experiment 13 requires approximately 107 seconds per epoch and completed 15 epochs in 26.7 minutes. Inference on 180 test images required about 15.3 minutes. Its measured training and inference peaks remain below the optional 5 GB and 4 GB thresholds.

8 Conclusions

The experiments show that ResNet18 transfer learning and preserving spatial detail are more useful than the specialised losses tested. Mean Dice and PQ measure different properties: a more selective prediction may recognise instances better while reducing overall mask overlap.

Experiment 13 is the delivered model: a ResNet18 U-Net trained on random 1536^2 crops from native 2048^2 frames. It obtains **0.6762 Mean Dice** and **0.3591 PQ**; peak GPU memory is 3.04 GB in training and 2.53 GB in inference.

The alternative 15m configuration reaches 0.6650 Mean Dice and 0.4153 PQ within the memory thresholds, while heavier models raise PQ but not Mean Dice at a much higher cost.

The main limitation is the possible optimism from reusing validation data for comparison and output selection; changing the training seed in some runs nevertheless produced nearly unchanged results. A useful continuation would improve recall for small filaments.